BIO102 Help Session - 2

Epistasis

Epistasis describes how interallelic gene interactions can affect phenotypes.

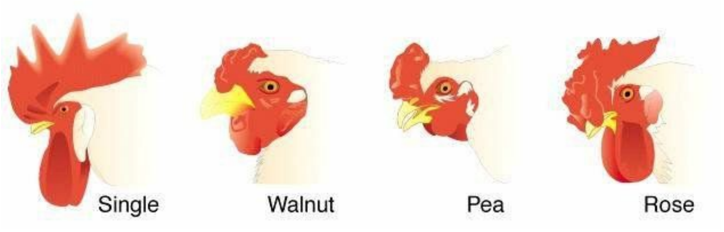
In Mendel's dihybrid cross, each gene locus had an independent effect on a single phenotype - two separate genes coded for two separate characteristics.

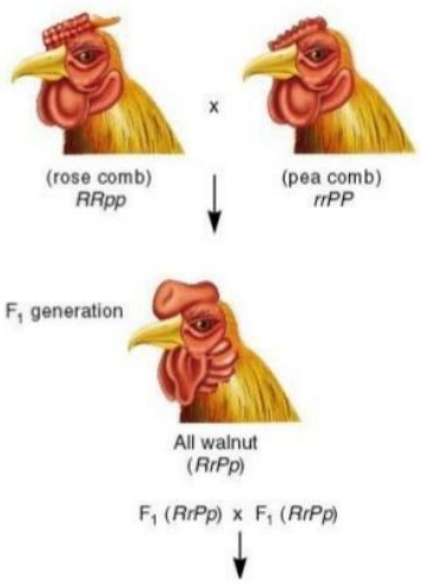
What happens when two different loci affect the same characteristic?

There are more than two gene products affecting the same phenotype, and these products may have complex relationships. Any time two different genes contribute to a single phenotype and their effects are not merely additive, those genes are said to be epistatic.

Epistasis is an interaction at the phenotypic level of organization. The genes that are involved in a specific epistatic interaction may still show independent assortment at the genotypic level. In such cases, however, the phenotypic ratios may appear to deviate from those expected with independent assortment.

Combs in Chickens





	RP	Rp	rP	rp
RP	RRPP Walnut	RRPp Walnut	RrPP Walnut	RrPp Walnut
Rp	RRPp Walnut	RRpp Rose	RrPp Walnut	Rrpp Rose
rP	RrPP Walnut	RrPp Walnut	rrPP Pea	rrPp Pea
rp	RrPp Walnut	Rrpp Rose	rrPp Pea	rrpp Single

Genotype	Ratio	Phenotype
R__P__	9	Walnut
R__pp	3	Rose
rrP___	3	Pea
rrpp	1	Single

Neither the rose comb nor the pea comb appeared to be dominant, because the F_1 offspring had their own unique phenotype.

When two of these F_1 progeny were crossed with each other, some of the members of the resulting F_2 generation had walnut combs, some had rose combs, some had pea combs, and some had a single comb. Because the four comb shapes appeared in a 9:3:3:1 ratio it seemed that two different genes must play a role in comb shape.

At R locus, R is dominant to r.
At P locus, P is dominant to p.
The loci are independently assorting.

Two pairs of genes determine the same phenotype but assorted independently, and the F2 phenotypic ratio 9 : 3 : 3 : 1 remains unaltered.

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Total Marks: 25

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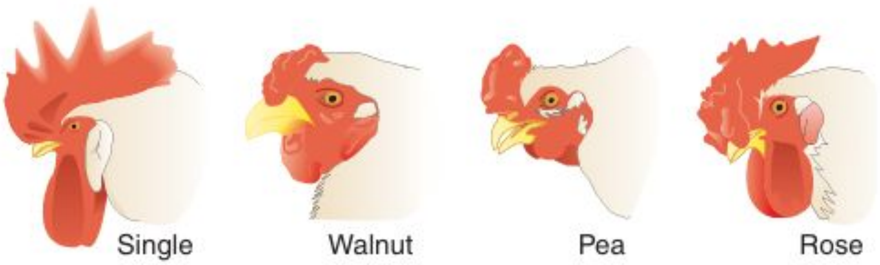
Name:

6. Observe the particular set of Drosophila crosses using pure breeding individuals of four eye colours- Orange-1; Orange- 2; Pink and Wild type. Write down the genotype of the four types of eye colour individuals. (3 marks)

Cross	F1 Phenotype
Wild-type X Orange-1	All wild-type
Wild-type X Orange-2	All wild-type
Orange-1 X Orange-2	All wild-type
Orange-2 X pink	All Orange-2
(F1 of (Orange-1 X Orange-2)) X pink	¼ Orange-2; ¼ pink; ¼ orange-1; ¼ wild-type

44. The genotype r/r ; p/p gives fowl a single comb, $R/-$; $P/-$ gives a walnut comb, r/r ; $P/-$ gives a pea comb, and $R/-$; p/p gives a rose comb (see the illustrations). Assume independent assortment.



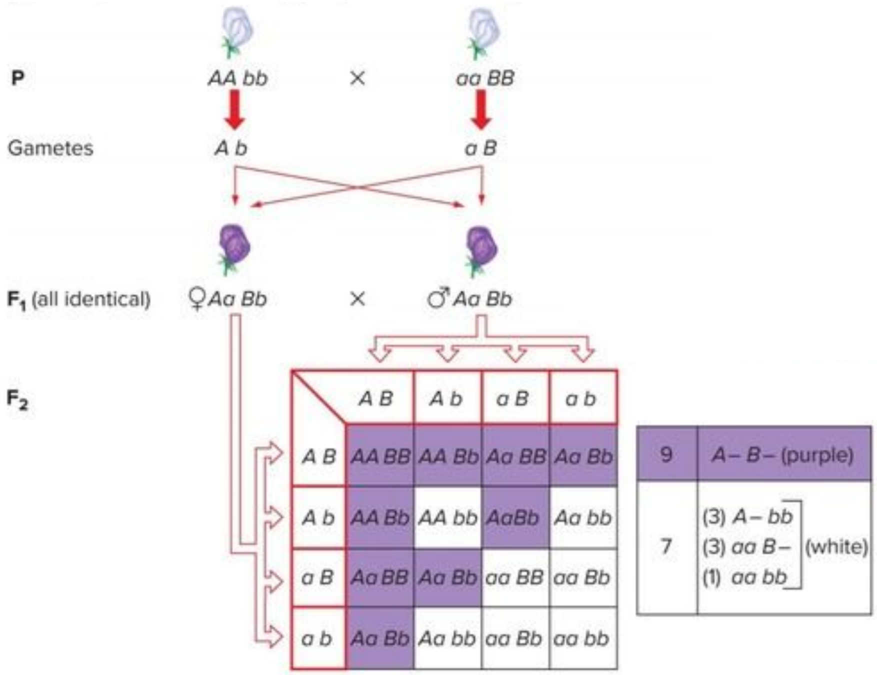
a. What comb types will appear in the F_1 and in the F_2 and in what proportions if single-combed birds are crossed with birds of a true-breeding walnut strain?

b. What are the genotypes of the parents in a walnut × rose mating from which the progeny are $\frac{3}{8}$ rose, $\frac{3}{8}$ walnut, $\frac{1}{8}$ pea, and $\frac{1}{8}$ single?

c. What are the genotypes of the parents in a walnut × rose mating from which all the progeny are walnut?

d. How many genotypes produce a walnut phenotype? Write them out.

Flower Colour in Sweet Pea



Two varieties of pea, each of which was pure-breeding for white flowers. This cross yielded an F_1 generation in which all progeny had purple flowers. Next, two F_1 plants were crossed to create the F_2 generation.

The ratio of purple flowers to white flowers was approximately 9:7.

Two recessive alleles at one flower locus could mask the effects of the alleles at the other flower locus.

The dominant alleles of two genes acting together (A– B–) produce color while the other three genotypic classes (A– bb, aa B–, and aa bb) do not.

If the sweet peas are either aa or bb, their flowers will be white regardless of whether or not they have a dominant allele of the other gene.

Any flower with the *aa* genotype would be white, no matter what alleles were present at its *B* locus.

Any flower with the *bb* genotype would also be white, no matter what alleles were present at its *A* locus.

"Epistasis" to define the masking action of one gene by another.

Primula Petal Colour - Malvidin Production

KKdd x kkDD

(blue) (white)

↓

KkDd

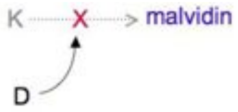
(selfed)

(white)

↓

♂/♀	KD	Kd	kD	kd
KD	KKDD	KKDd	KkDD	KkDd
Kd	KKDd	KKdd	KkDd	Kkdd
kD	KkDD	KkDd	kkDD	kkDd
kd	KkDd	Kkdd	kkDd	kkdd

Pigment malvidin creates blue-colored flowers.



Synthesis of malvidin is controlled by gene *K*, yet production of this pigment can be suppressed by gene *D*, which is found at completely different locus

In this case, the *D* allele is dominant to the *K* allele, so plants with the genotype *KkDd* will not produce malvidin because of the presence of the *D* allele

Phenotypic ratio - 13:3

K_D_ : 9
K_dd : 3
kkD_ : 3
kkdd : 1

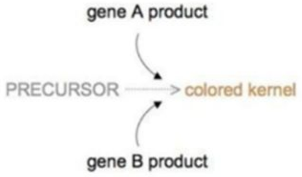
This type of epistasis is sometimes called dominant suppression, because the deviation from 9:3:3:1 is caused by a single allele that produces a dominant phenotype, and the action of this allele is to suppress the expression of some other gene.

Wheat Kernel Colour

AaBb x AaBb

		Female Gametes			
		AB	*Ab*	*aB*	*ab*
Male Gametes	*AB*	*AABB*	*AABb*	*AaBB*	*AaBb*
	Ab	*AABb*	*AAbb*	*AaBb*	*Aabb*
	aB	*AaBB*	*AaBb*	*aaBB*	*aaBb*
	ab	*AaBb*	*Aabb*	*aaBb*	*aabb*

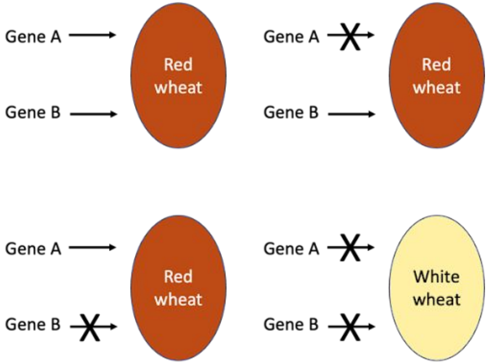
Genotype	Phenotype	Enzymatic Activites	
9 A_B_	Colored Kernels	Functional enzymes from both genes	15
3 A_bb	Colored Kernels	Functional enzymes from A genes	
3 aaB_	Colored Kernels	Functional enzymes from B genes	
1 aabb	White Kernels	Non Functional enzymes from both genes	1



Kernel color is dependent upon a biochemical reaction that converts a precursor substance into a pigment, and this reaction can be performed with the product of either gene *A* or gene *B*

Thus, having either an *A* allele or a *B* allele produces color in the kernel, but a lack of either allele will produce a white kernel that is devoid of color.

In this cross, whenever a dominant allele is present at either locus, the biochemical conversion occurs, and a colored kernel results. Thus, only the double homozygous recessive genotype produces a phenotype with no color, and the resulting phenotypic ratio of color to non-color is 15:1.



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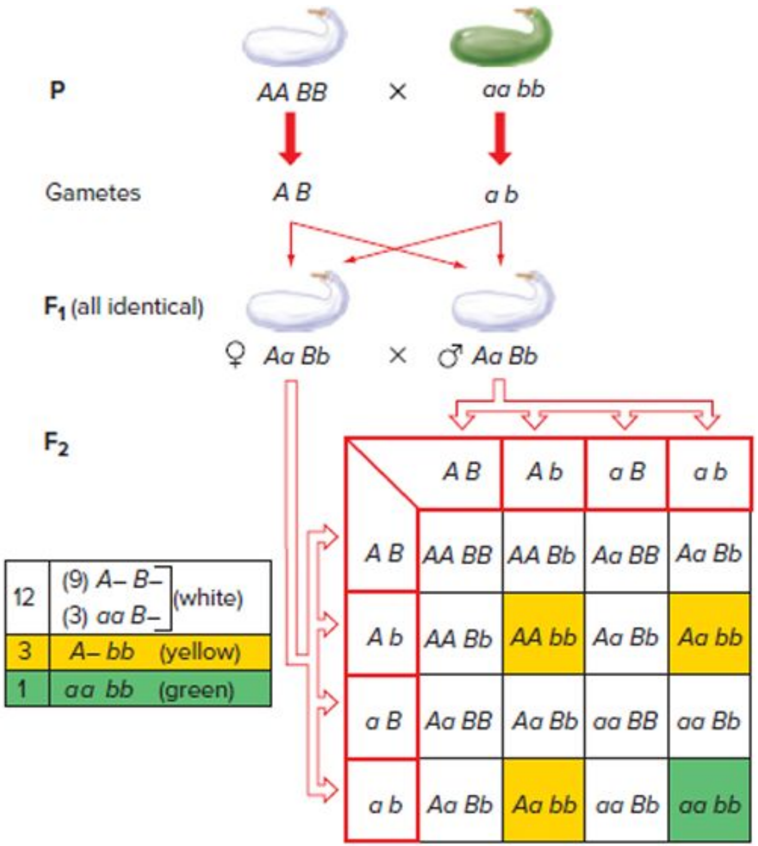
3. Assume that in a dihybrid cross starting from pure breeding parents, the F2 shows a 13:3 ratio for two phenotypes. If we take the F1 individual from the above cross and subject it to a test cross, what is the expected phenotypic ratio? Explain. (3 marks)

Summer Squash Colour



In summer squash, two genes influence the color of the fruit.

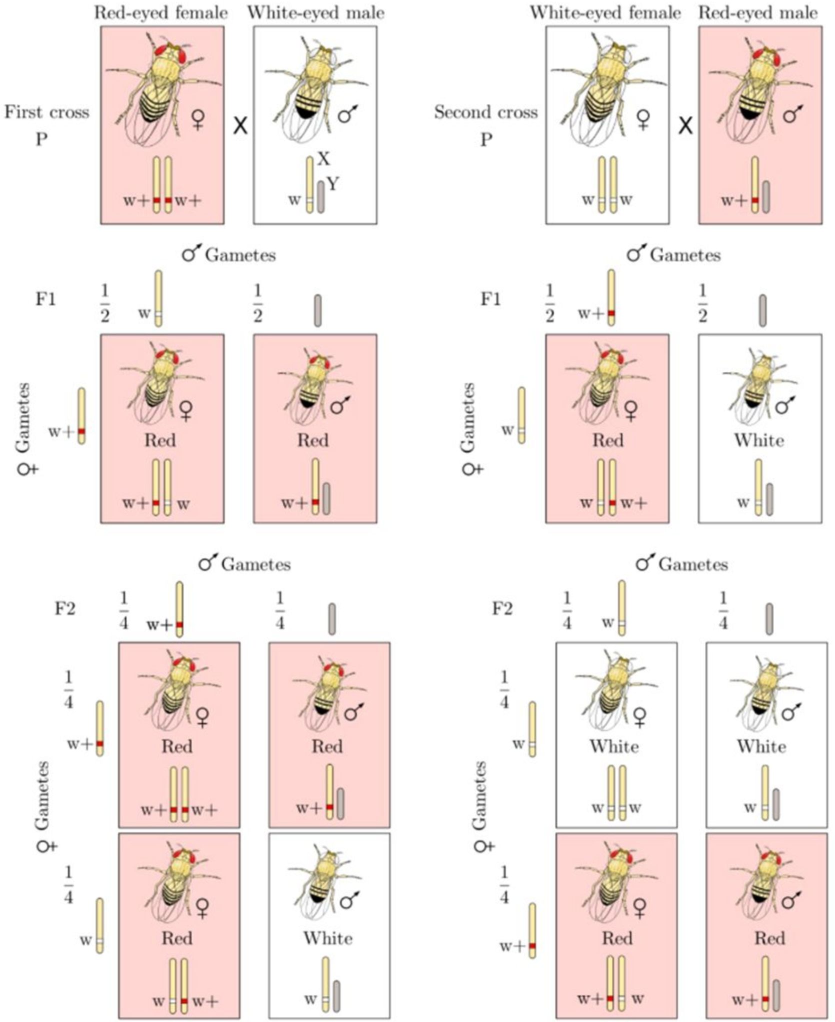
With one gene, the dominant allele (A–) determines yellow, while homozygotes for the recessive allele (aa) are green.

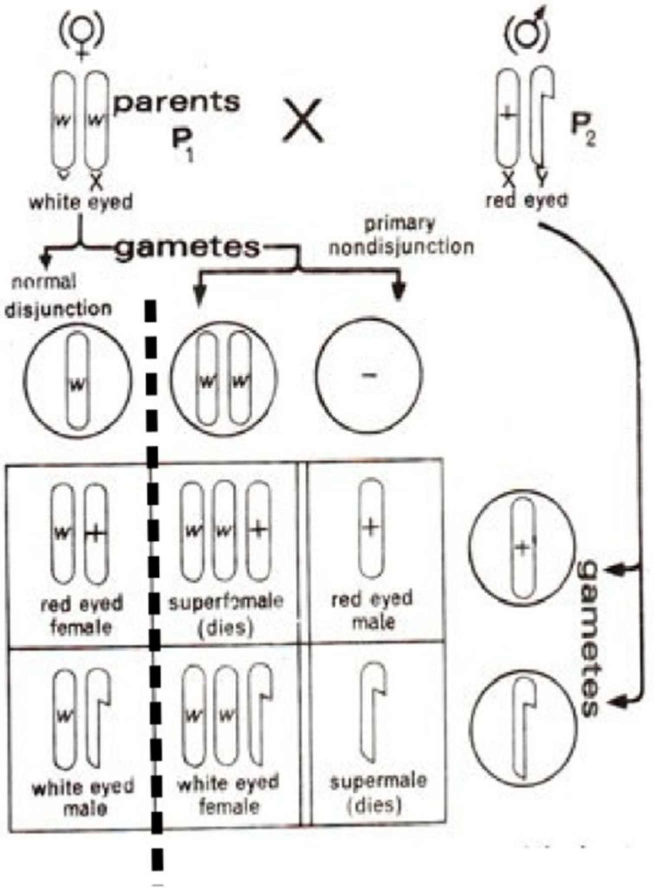
A second gene's dominant allele (B–) produces white, while bb fruit may be either yellow or green, depending on the genotype of the first gene.

In the interaction between these two genes, the presence of B hides the effects of either A– or aa, producing white fruit, and B– is thus epistatic to any genotype of the A gene.

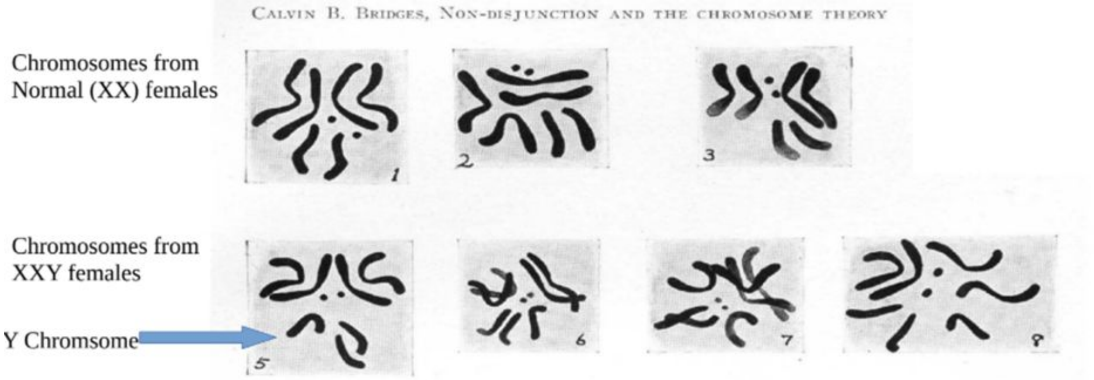
The recessive b allele has no effect on fruit color determined by gene A.

Epistasis in which the dominant allele of one gene hides the effects of another gene is called dominant epistasis.


Red-eyed female
White-eyed male
First cross
P
♀
X
♂
X
Y
w+
w+
w
♂ Gametes
F1
1/2
w
1/2
♀ Gametes
w+
Red
♀
w+
w
Red
♂
w+
♂ Gametes
F2
1/4
w+
1/4
♀ Gametes
1/4
w+
Red
♀
w+
w+
Red
♂
w+
1/4
w
Red
♀
w
w+
White
♂
w
White-eyed female
Red-eyed male
Second cross
P
♀
X
♂
w
w
w+
♂ Gametes
F1
1/2
w+
1/2
♀ Gametes
w
Red
♀
w
w+
White
♂
w
♂ Gametes
F2
1/4
w
1/4
♀ Gametes
1/4
w
White
♀
w
w
White
♂
w
1/4
w+
Red
♀
w+
w
Red
♂
w+



parents
P_1
X
P_2
white eyed
red eyed
gametes
normal disjunction
primary nondisjunction
red eyed female
superfemale (dies)
red eyed male
white eyed male
white eyed female
supermale (dies)
gametes



Calvin B. Bridges, Non-disjunction and the chromosome theory
Chromosomes from Normal (XX) females
Chromosomes from XXY females
Y Chromsome

7. Based on the following Drosophila crosses, explain the genetic basis for each trait and determine the genotypes of all individuals:

white-eyed, dark-bodied female X red-eyed, tan-bodied male

F 1 : females are all red-eyed, tan-bodied; males are all white-eyed, tan-bodied

F 2 : 27 red-eyed, tan-bodied
24 white-eyed, tan-bodied
9 red-eyed, dark-bodied
7 white-eyed, dark-bodied

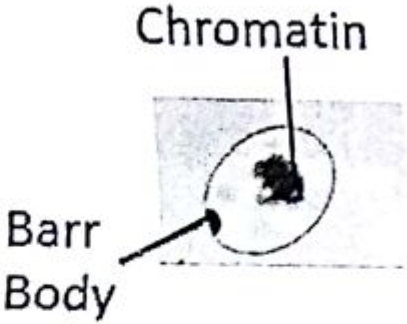
(No differences between males and females in the F 2 generation.)

2. A white-eyed male fly is mated with a pink-eyed female. All the F1 offspring have wild-type red eyes. F1 individuals are mated among themselves to yield:

Females	Males
red-eyed 450	red-eyed 231
pink-eyed 155	white-eyed 301
	pink-eyed 70

Provide a genetic explanation for the results. (4 marks)

1. The photograph in the margin is of a human cell nucleus stained with a suitable dye. Can you guess the sex of the individual from whom the cell is taken? Explain. (2 marks)



9. How many Barr bodies would you see in the nuclei of persons with the following sex chromosomes?
a. X0
b. XXX
c. XX
d. XXXXX
e. XY
f. XXY

3. Calico cats have large patches of colored fur. Tortoiseshell cats have very small color patches. Both these phenotypes are supposed to be because of random X chromosome inactivation (also called Lyonisation). Explain the difference between the two phenotypes based on the age of onset of Lyonization (is it early or late during embryonic development). (3)